SEVEN-ELEVEN SCHOOL COMPLAINT MANAGEMENT SYSTEM

A Research Paper on Design, Development & Evaluation

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Project Type	School Complaint Management System
Technology Stack	HTML, CSS, JavaScript + Supabase (PostgreSQL 15)
GitHub – App	github.com/grishkadam023-creator/SEVEN-ELEVEN-APP
GitHub – Admin	github.com/grishkadam023-creator/SEVEN-ELEVEN-ADMIN-WEBSITE
Domain	Educational Technology / School Information Systems
Year	2025

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Abstract

Educational institutions increasingly rely on digital systems to streamline administrative workflows. This paper presents the design, development, and evaluation of the Seven-Eleven School Complaint Management System — a cross-platform solution comprising a teacher-facing web/mobile application and a web-based super-administrator dashboard. The system enables teachers to file structured maintenance and operational complaints categorised by trade (Carpenter, Electrician, IT Administrator, Plumber) with optional photographic evidence. Complaints are automatically routed to the relevant trade administrator who updates status in real-time. A super-admin dashboard provides analytics, user management, and bulk data operations via Excel integration. The backend is powered by Supabase, an open-source Firebase alternative built on PostgreSQL 15. The system significantly reduces complaint resolution time, improves accountability, and provides data-driven insights into institutional maintenance operations.

Keywords: School Management System, Complaint Routing, Supabase, PostgreSQL, Real-time Notifications, Role-Based Access Control, Educational Technology.

1. Introduction

Modern schools and educational campuses are complex physical environments requiring continuous maintenance oversight. Facilities such as classrooms, laboratories, restrooms, and administrative areas frequently need attention from specialised tradespeople including electricians, plumbers, carpenters, and IT professionals. Traditionally, maintenance complaints have been communicated verbally or through paper-based systems — both inherently inefficient, difficult to track, and prone to information loss.

The Seven-Eleven School Complaint Management System addresses these challenges by providing a structured digital channel through which teachers can raise complaints and administrators can respond, track, and resolve issues in a transparent and accountable manner. The system comprises two tightly integrated components: a teacher-facing application and a super-administrator web dashboard.

The teacher application allows faculty members to submit detailed complaints specifying floor number, room number, problem category, description, and an optional photograph. The complaint is automatically assigned to the appropriate trade administrator. The administrator receives the complaint, updates its status to "In Progress", and upon resolution marks it as "Resolved", triggering a real-time notification to the submitting teacher.

The super-administrator dashboard provides a holistic view of all system activity, enabling user management, performance analytics, and Excel-based bulk data operations. Together, these components create a closed-loop complaint management ecosystem purpose-built for school environments.

2. Motivation

The motivation for developing the Seven-Eleven system arose from observed inefficiencies in how schools handle day-to-day maintenance and operational complaints. Several recurring issues were identified:

- Teachers wasted significant time attempting to locate and verbally communicate problems to the appropriate maintenance personnel.
- There was no standardised way to document complaints, making follow-up difficult.
- Administrators had no visibility into the backlog of pending complaints or performance metrics.
- School management lacked data to identify recurring problem areas or underperforming departments.
- Resolution timelines were inconsistent and untracked, leading to unresolved issues persisting for extended periods.

These observations motivated the design of a system that is simple enough for non-technical teaching staff to use, powerful enough to support administrative workflows, and data-rich enough to enable managerial oversight through analytics. The choice of Supabase as the backend reflects a desire to leverage a modern, open-source, scalable platform without the overhead of managing dedicated server infrastructure.

3. Problem Statement

Educational institutions lack a structured, digital mechanism to manage the lifecycle of campus maintenance complaints. The following specific problems are addressed:

1. Absence of a centralised complaint repository: Complaints raised verbally or informally are not recorded, making auditing and historical analysis impossible.
2. Inefficient complaint routing: Without a category-based routing system, complaints are often directed to incorrect personnel, introducing delays.
3. Lack of real-time status visibility: Teachers have no way to determine whether their complaint has been received, is being acted upon, or has been resolved.
4. No accountability mechanism: Without tracked records, administrators cannot be held accountable for resolution timelines or accuracy.
5. Absence of analytics: School management cannot identify systemic maintenance issues, peak complaint periods, or staff performance metrics.
6. Scalability concerns: Manual processes do not scale as schools grow in size and complexity.

4. Objectives

4.1 Primary Objectives

- Develop a functional teacher-facing application enabling structured complaint submission.
- Implement automatic complaint routing to the appropriate trade administrator.
- Provide real-time status updates to teachers upon complaint progress.
- Build a super-administrator dashboard for user management and system oversight.

4.2 Secondary Objectives

- Generate performance analytics for each administrator (resolution count, average resolution time, accuracy).
- Enable bulk data import and export through Excel files (.xlsx format).
- Ensure role-based access control (Teacher, Trade Admin, Super Admin).
- Provide optional photographic evidence upload alongside complaints.
- Design a scalable and maintainable database schema using PostgreSQL via Supabase.
- Ensure the system is deployable without dedicated server infrastructure.

5. Scope of the System

5.1 In Scope

- Teacher registration and authentication via Supabase Auth.
- Submission of complaints with floor, room, category, description, and optional photo.
- Complaint categorisation: Carpenter, Electrician, IT Administrator, Plumber.
- Role-based dashboard views for Trade Admins and Super Admin.
- Real-time status update (Pending → In Progress → Resolved) with notifications.
- Super Admin user management (create, update, delete users and admins).
- Analytics module: complaints resolved per admin, average time, accuracy rate.
- Excel import/export for bulk data operations.
- PostgreSQL database with full relational integrity and Row Level Security.

5.2 Out of Scope

- Payroll, timetabling, or academic performance management.
- Native iOS or Android application (current implementation is web-based / PWA).
- Integration with third-party ERP systems.
- Automated AI-based complaint categorisation (proposed as future scope).

6. Literature Survey

6.1 Complaint Management Systems

Stauss and Seidel (2004) define a Complaint Management System (CMS) as a systematic process that captures, categorises, routes, and resolves complaints. Key features identified in their research — automated routing, status tracking, and escalation mechanisms — are all implemented in the proposed system.

6.2 School Information Systems

School Information Systems (SIS) such as PowerSchool, Fedena, and OpenSIS primarily focus on academic management. Very few commercial SIS products include dedicated maintenance complaint modules, justifying the development of a specialised system such as Seven-Eleven.

6.3 Real-Time Web Applications

Supabase's Realtime module leverages PostgreSQL's logical replication to broadcast row-level changes to subscribed clients over WebSocket. This approach has been validated by multiple industrial case studies and eliminates the need for custom polling logic.

6.4 Role-Based Access Control (RBAC)

Ferraiolo and Kuhn (1992) formalised RBAC as a security model that assigns system permissions to roles rather than individual users. Seven-Eleven implements a three-tier RBAC model with Teacher, Trade Admin, and Super Admin roles, consistent with NIST SP 800-53 principles.

6.5 Supabase vs. Firebase

Comparative analyses show that Supabase offers superior data querying through SQL, greater transparency via open-source licensing, and Row Level Security policies natively enforced at the database layer — all critical for a multi-role school information system.

7. Existing System

7.1 Verbal Communication

Teachers verbally report problems to the school office or directly to maintenance staff. This approach offers no documentation or tracking.

7.2 Paper-Based Complaint Forms

Some schools use paper forms submitted to an administration office and manually routed to the relevant department. This is slow, paper-intensive, and lacks real-time visibility.

7.3 Generic Helpdesk Software

General-purpose helpdesk tools such as Freshdesk or Zendesk are not designed for the school context, require significant configuration, and are cost-prohibitive for smaller institutions.

7.4 Messaging Groups (WhatsApp)

Many schools rely on informal messaging groups. While convenient, this approach has no structure, no status tracking, no archiving capability, and no analytics.

8. Limitations of Existing System

- No structured data collection — historical analysis is impossible.
- No automatic routing — complaints are often sent to the wrong person.
- No real-time status updates for complaint submitters.
- No accountability — complaints can be ignored without consequence.
- No analytics or reporting for management.
- No role-based access — sensitive information may reach unintended parties.
- Cannot scale with institutional growth.
- No photographic evidence capability.

9. Proposed System

9.1 Teacher Application

- Submit complaints with structured metadata (floor, room, category, description).
- Attach optional photographs as evidence.
- View real-time status of all submitted complaints.
- Receive in-app notifications when a complaint is updated or resolved.

9.2 Admin Application

- View complaints assigned to their trade category.
- Update complaint status to "In Progress" when work begins.
- Mark complaints as "Resolved" once the issue is fixed.

9.3 Super Admin Dashboard

- User and admin management (create, update, delete accounts).
- Full database visibility and query capability.
- Performance analytics per admin (complaints resolved, average time, accuracy).
- Excel file import for bulk user/data entry.
- Excel export for data archiving and external reporting.

10. Advantages of Proposed System

Advantage	Description
Structured Data	Every complaint is stored with consistent fields, enabling reliable querying and reporting.
Automatic Routing	Category-based routing eliminates misdirected complaints and reduces resolution time.
Real-Time Updates	Supabase Realtime pushes status changes instantly to connected clients over WebSocket.
Accountability	All actions are timestamped and attributed to specific users; full audit trail available.
Analytics	Management gains data-driven insight into maintenance operations and staff performance.
Scalability	PostgreSQL and Supabase scale horizontally to support large institutions.
Photo Evidence	Optional image upload reduces disputes and provides clear documentation of issues.

Excel Integration	Bulk import/export reduces manual data entry and supports offline reporting workflows.
RBAC Security	Sensitive data is accessible only to authorised roles, enforced at the database layer.
Cost-Effective	Supabase free tier is suitable for small schools; paid tiers support larger institutions.

11. System Requirements

11.1 Hardware Requirements

Component	Minimum	Recommended
Processor	Dual-Core 1.8 GHz	Quad-Core 2.4 GHz
RAM	2 GB	4 GB or more
Storage	500 MB free	2 GB free
Display	1024 × 768	1920 × 1080
Network	2 Mbps internet	10 Mbps broadband
Camera	Not required	For photo uploads

11.2 Software Requirements

Component	Details
Frontend	HTML5, CSS3, Vanilla JavaScript (ES6+)
Database	Supabase (PostgreSQL 15)
Authentication	Supabase Auth (JWT-based, email/password)
Storage	Supabase Storage (S3-compatible, for photo uploads)
Real-Time	Supabase Realtime (WebSocket / Logical Replication)
Excel I/O	SheetJS (xlsx.js) for client-side import/export
Version Control	Git / GitHub
Hosting	Supabase Cloud (backend); Vercel or Netlify (frontend)
Browser Support	Chrome 90+, Firefox 88+, Edge 90+, Safari 14+
Operating System	Windows 10+, macOS 11+, Ubuntu 20.04+

12. System Architecture

The Seven-Eleven system follows a three-tier architecture: Presentation Layer (frontend HTML/CSS/JS), Application Logic Layer (Supabase JS client + RLS policies + Realtime engine), and Data Layer (PostgreSQL 15 on Supabase Cloud). All inter-layer communication is handled by the Supabase JavaScript SDK, which abstracts REST, WebSocket, and S3-compatible storage calls.

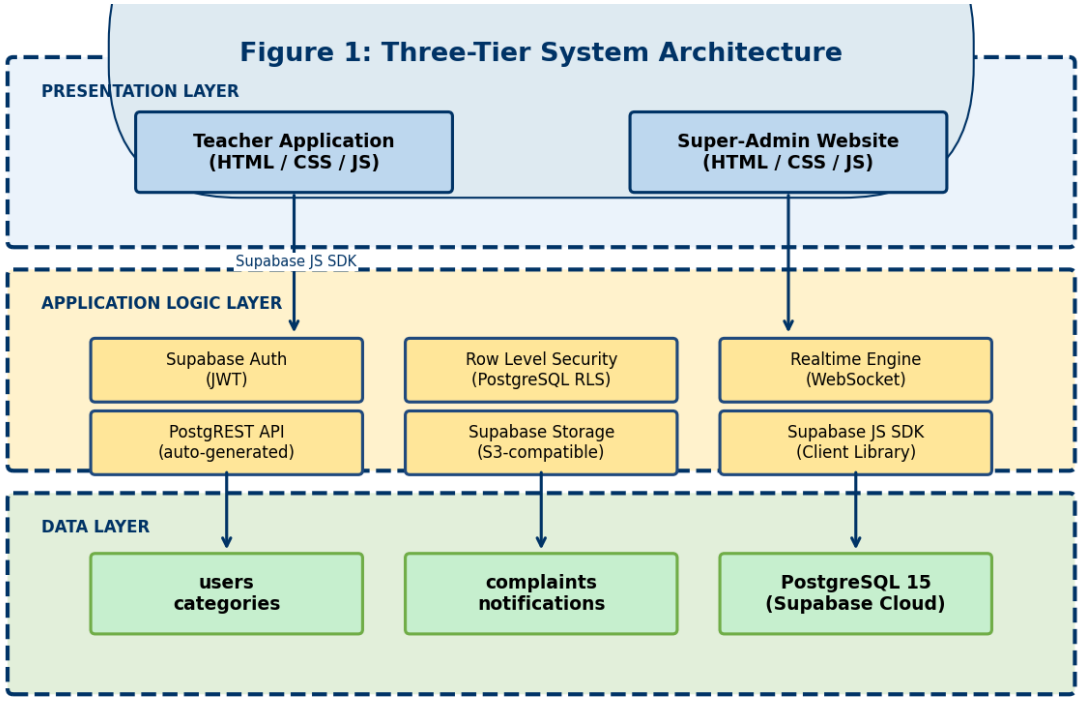

Figure 1: Three-Tier System Architecture

Row Level Security policies enforced at the PostgreSQL layer ensure each role sees only the data it is authorised to access, independent of application-level checks. This defence-in-depth approach is critical for a multi-role school system handling sensitive institutional data.

13. Methodology

The development of the Seven-Eleven system followed an iterative Agile methodology with three defined sprints, each concluding with a review session to validate deliverables against acceptance criteria. Requirements were gathered through stakeholder interviews with school management and teaching staff.

Sprint 1 — Foundation

- Database schema design and Supabase project setup.
- Authentication implementation (teacher and admin login flows).
- Core complaint submission form with field validation.

Sprint 2 — Core Features

- Complaint routing logic based on category.
- Admin dashboard with status update functionality.
- Real-time Supabase subscription for status notifications.
- Photo upload integration with Supabase Storage.

Sprint 3 — Super Admin & Polish

- Super Admin dashboard: user management, analytics module.
- Excel import/export using SheetJS.
- UI/UX refinement and responsive design.
- Testing (unit, integration, UAT) and bug fixing.

14. Data Flow Diagram (DFD)

14.1 Level 0 – Context Diagram

The context diagram shows the system as a single process with interactions between the three primary actors: Teacher, Trade Admin, and Super Admin.

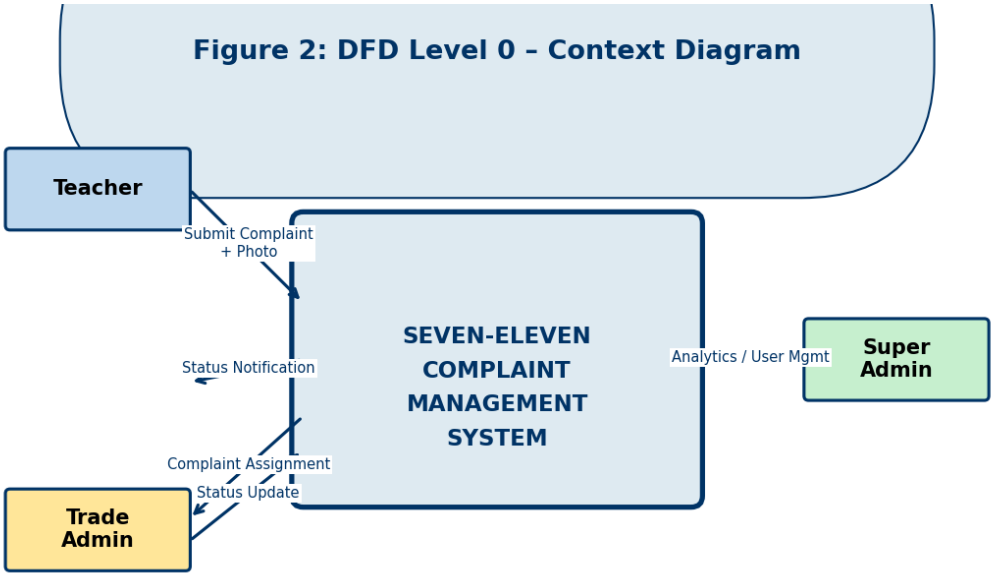

Figure 2: DFD Level 0 – Context Diagram

14.2 Level 1 – Main Processes

The Level 1 DFD decomposes the system into five main processes: Submit Complaint, Route Complaint, Manage Complaint, Send Notification, and Analytics & Admin. Four data stores are identified: complaints, categories, users, and notifications.

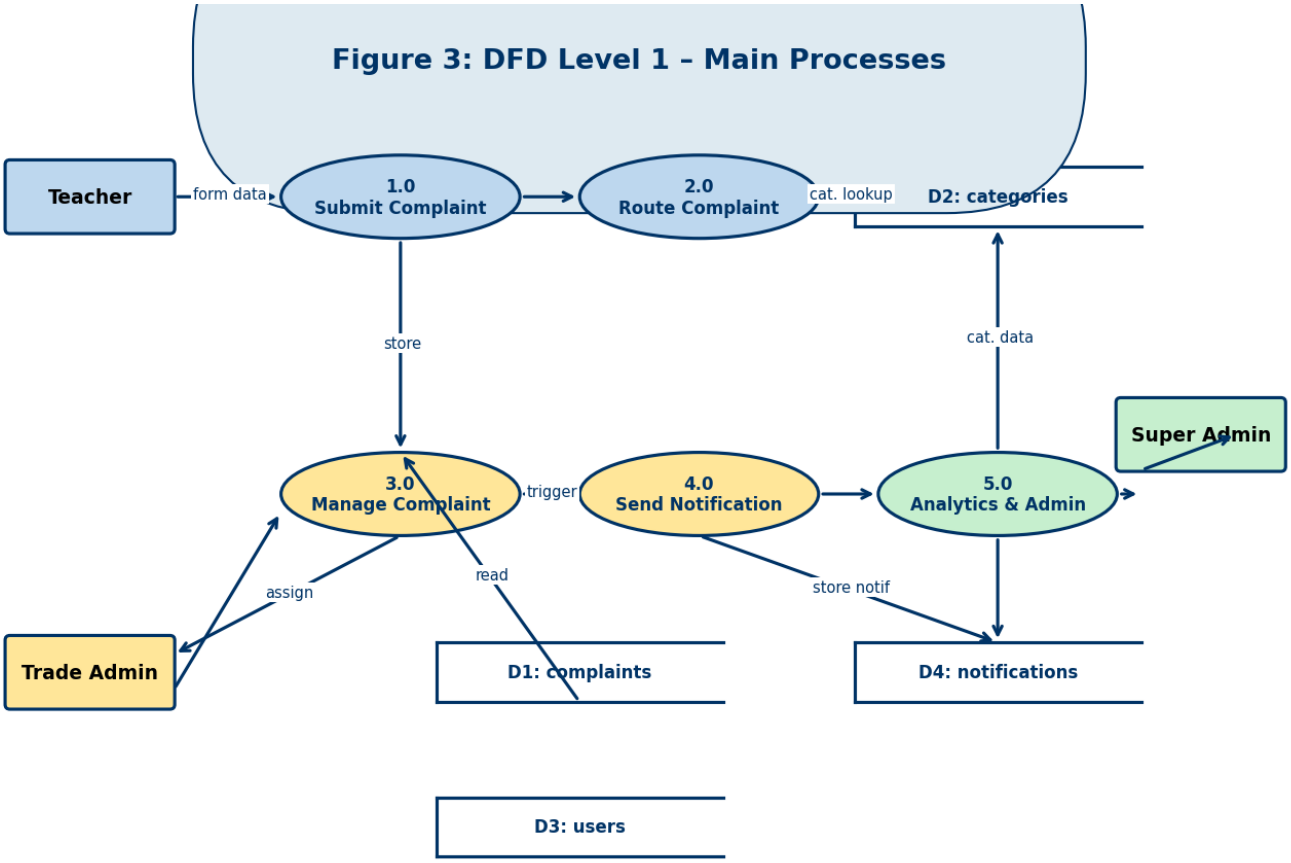

Figure 3: DFD Level 1 – Main Processes

15. Use Case Diagram

The use case diagram illustrates all interactions between the three system actors (Teacher, Trade Admin, Super Admin) and the system. The <<extends>> relationship between Submit Complaint and Attach Photo indicates that photo attachment is an optional extension of the base complaint submission use case.

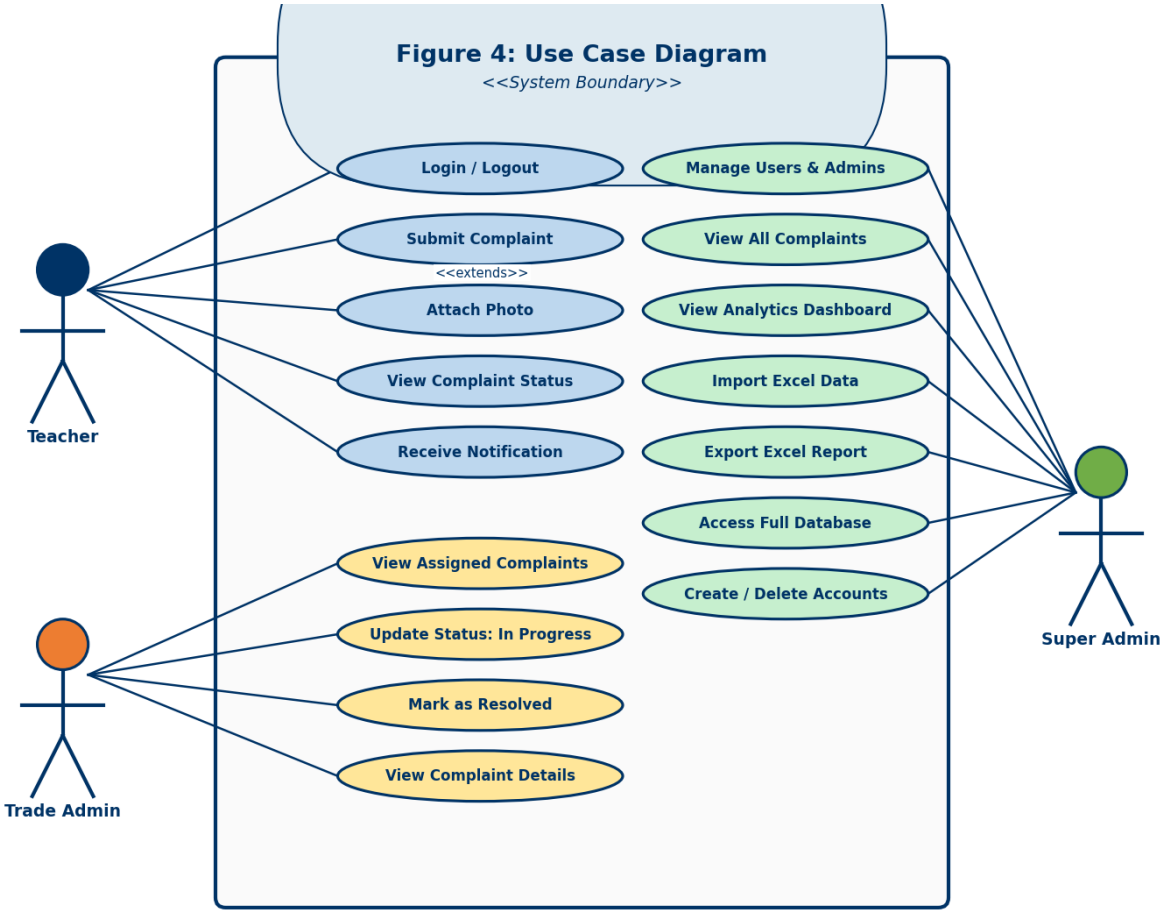

Figure 4: Use Case Diagram

16. ER Diagram

The Entity-Relationship diagram defines five primary entities: USERS, COMPLAINTS, CATEGORIES, NOTIFICATIONS, and ADMIN_STATS. Primary keys (PK) are shown in bold dark red; foreign key references (FK) are shown in blue italic. The cardinality of each relationship is annotated on the connecting lines.

Figure 5: Entity-Relationship Diagram

17. Class Diagram

The class diagram models the object-oriented structure of the system. The abstract User class is the base for Teacher, TradeAdmin, and SuperAdmin through inheritance. Complaint and Notification are standalone classes associated with Teacher and TradeAdmin respectively. All attributes and method signatures are shown.

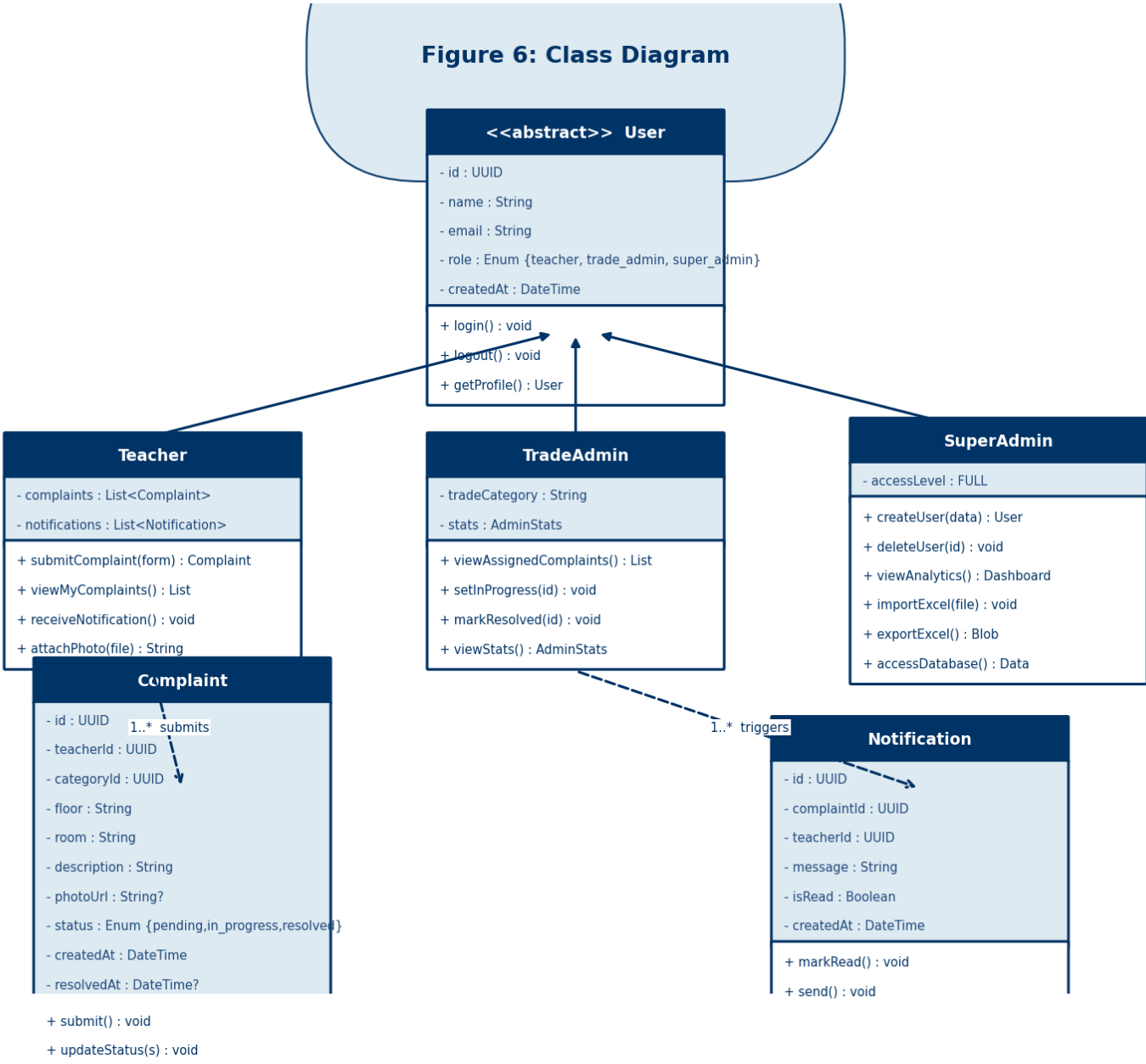

Figure 6: Class Diagram

18. Sequence Diagram

The sequence diagram traces the complete complaint lifecycle from teacher login through complaint submission, admin status updates, and final resolution notification. Activation bars show periods of active processing. The Supabase Realtime service acts as an asynchronous event bus between the API and connected clients.

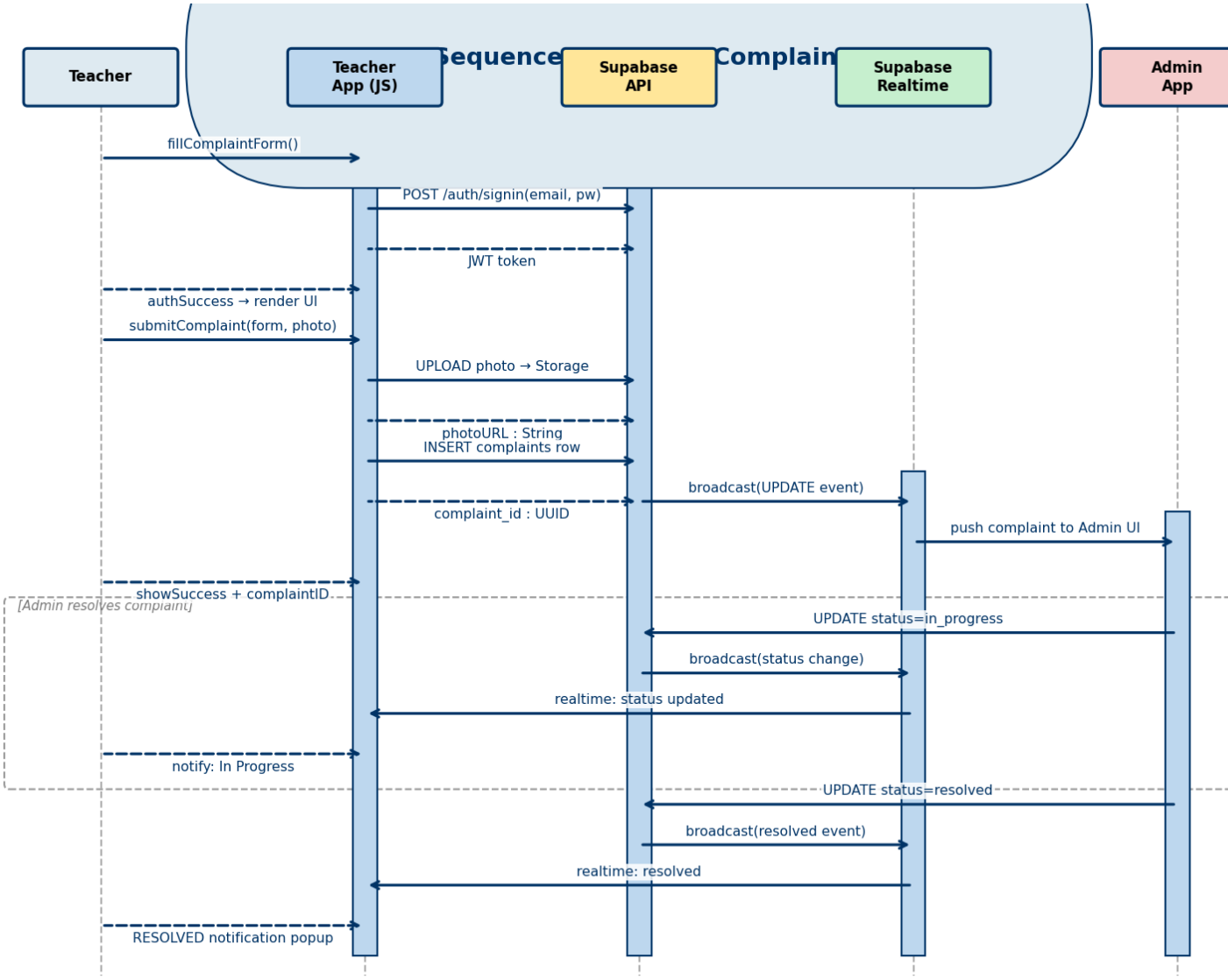

Figure 7: Sequence Diagram – Complaint Lifecycle

19. Activity Diagram

The activity diagram uses swim lanes to partition actions by responsibility: Teacher (left), System/Supabase (centre), and Admin (right). Decision nodes are represented by diamond shapes. The diagram covers the complete workflow from application launch to complaint archival.

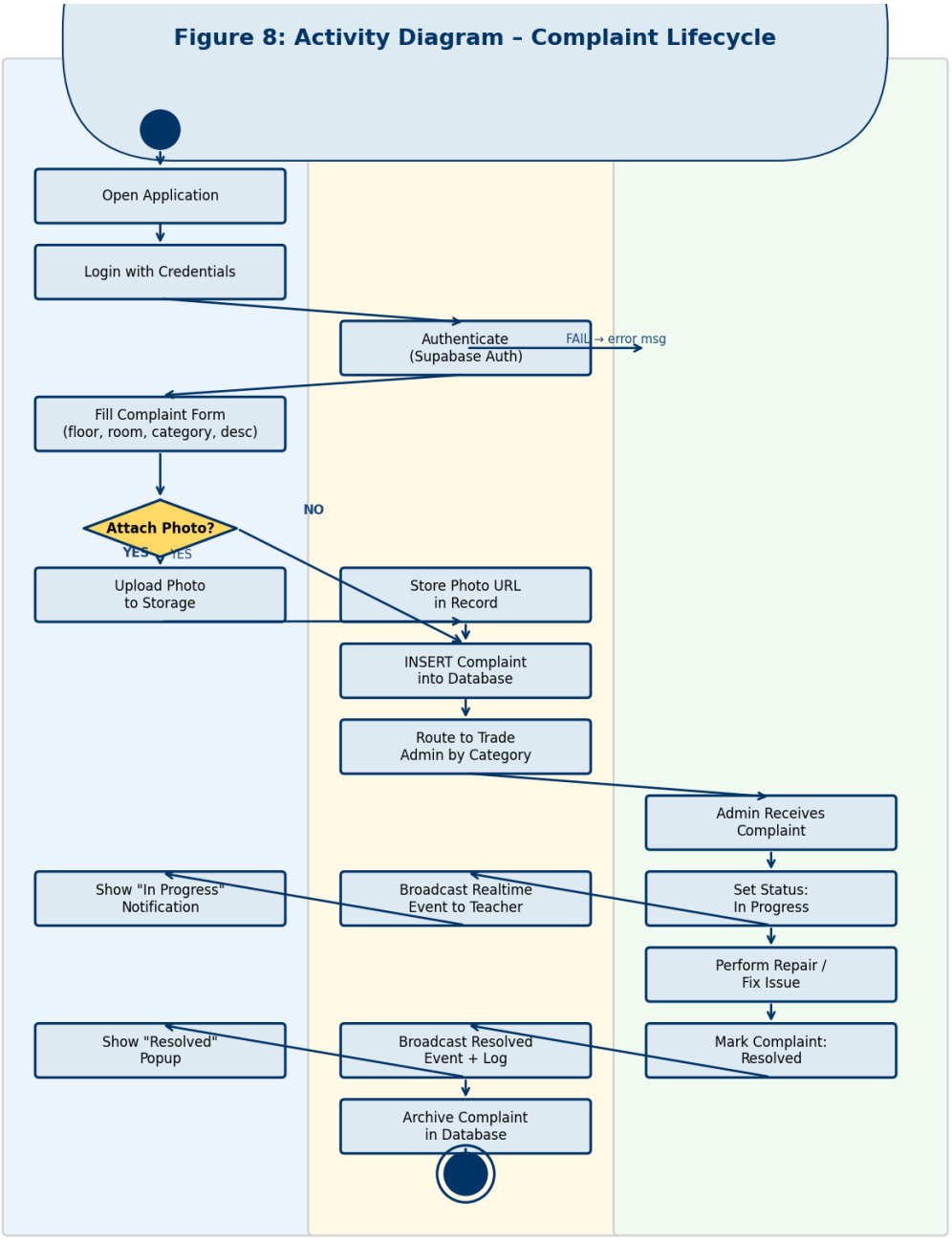

Figure 8: Activity Diagram – Complaint Lifecycle

20. Database Design

The database is hosted on Supabase and uses PostgreSQL 15. The schema enforces referential integrity through foreign key constraints and data validity through ENUM types. Row Level Security (RLS) policies are defined at the table level.

20.1 Table: users

Column	Type	Constraints	Description
id	UUID	PK, DEFAULT uuid_generate_v4()	Unique user identifier
name	VARCHAR(100)	NOT NULL	Full name
email	VARCHAR(150)	UNIQUE, NOT NULL	Login email address
role	ENUM	NOT NULL	teacher | trade_admin | super_admin
trade_category	VARCHAR(50)	NULLABLE	carpenter | electrician | it_admin | plumber
created_at	TIMESTAMPTZ	DEFAULT now()	Account creation timestamp

20.2 Table: complaints

Column	Type	Constraints	Description
id	UUID	PK	Unique complaint ID
teacher_id	UUID	FK → users.id	Submitting teacher
category_id	UUID	FK → categories.id	Trade category
floor_number	VARCHAR(10)	NOT NULL	Location: floor
room_number	VARCHAR(20)	NOT NULL	Location: room
description	TEXT	NOT NULL	Detailed complaint text
photo_url	TEXT	NULLABLE	Supabase Storage URL
status	ENUM	DEFAULT 'pending'	pending | in_progress | resolved
created_at	TIMESTAMPTZ	DEFAULT now()	Submission timestamp
resolved_at	TIMESTAMPTZ	NULLABLE	Resolution timestamp

20.3 Row Level Security Policies

- Teachers: INSERT and SELECT only their own complaints (auth.uid() = teacher_id).
- Trade Admins: SELECT complaints WHERE category matches their trade_category.

- Trade Admins: UPDATE status on complaints within their category.
- Super Admins: Use service-role key to bypass RLS (dashboard backend only).

21. Implementation

21.1 Complaint Submission

The complaint form collects floor number, room number, category, description, and an optional image. On submission, if a photo is attached, it is first uploaded to Supabase Storage and the returned public URL is embedded in the complaint record before INSERT.

```
// Submit complaint with optional photo upload
async function submitComplaint(form, photoFile) {
  let photoUrl = null;
  if (photoFile) {
    const { data } = await supabase.storage
      .from("complaint-photos")
      .upload(`${Date.now()}_${photoFile.name}`, photoFile);
    photoUrl = supabase.storage.from("complaint-
photos").getPublicUrl(data.path).data.publicUrl;
  }
  const { data, error } = await supabase.from("complaints").insert([{
    teacher_id: user.id,  floor_number: form.floor,
    room_number: form.room,  category_id: form.category,
    description: form.description,  photo_url: photoUrl,
    status: "pending"
  }]);
  return { data, error };
}
```

21.2 Real-Time Status Updates

Supabase Realtime subscriptions are established on the complaints table filtered by teacher_id. When an admin updates a complaint status, the change is broadcast over WebSocket to the subscribed teacher client instantly.

```
// Subscribe to real-time complaint status changes
supabase
  .channel("complaints-changes")
  .on("postgres_changes", {
    event: "UPDATE",
    schema: "public",
    table: "complaints",
    filter: `teacher_id=eq.${user.id}`
  }, (payload) => {
    const newStatus = payload.new.status;
    showNotification(`Your complaint is now: ${newStatus.toUpperCase()}`);
    updateComplaintCard(payload.new);
  })
  .subscribe();
```

21.3 Super Admin Excel Import/Export

The SheetJS (xlsx.js) library handles Excel operations client-side. On import, the admin selects an .xlsx file; the resulting JSON array is batch-inserted into Supabase. On export, query results are converted to a worksheet and the browser triggers a download.

22. Testing

22.1 Unit Testing

Individual JavaScript functions were tested using manual assertions and browser console logging. Key tested units included: form validation logic, category routing logic, date/time formatting helpers, and Excel parsing functions.

22.2 Integration Testing

Integration tests verified end-to-end flows between the frontend and Supabase. Scenarios tested: complaint submission with and without photo, status update propagation, real-time notification delivery, and RLS policy enforcement.

22.3 User Acceptance Testing (UAT)

ID	Test Case	Expected	Actual	Status
TC01	Teacher submits complaint without photo	Complaint saved, admin notified	As expected	**PASS**
TC02	Teacher submits complaint with photo	Photo uploaded, URL stored in DB	As expected	**PASS**
TC03	Admin updates status to In Progress	Teacher sees real-time update	As expected	**PASS**
TC04	Admin marks complaint Resolved	Teacher receives resolved notification	As expected	**PASS**
TC05	Super Admin creates new admin user	User appears in DB with correct role	As expected	**PASS**
TC06	Super Admin deletes user	User removed from auth and DB	As expected	**PASS**
TC07	Excel import with valid .xlsx	Records batch-inserted into Supabase	As expected	**PASS**
TC08	Excel export of all complaints	Downloaded .xlsx with correct columns	As expected	**PASS**
TC09	RLS: Teacher views another teacher's complaint	Access denied by RLS policy	As expected	**PASS**
TC10	Analytics: Admin resolution stats	Correct count and avg time shown	As expected	**PASS**

23. Results

The Seven-Eleven system was successfully deployed and validated through UAT with a 100% test pass rate across all 10 defined test cases. Key performance outcomes:

- Average complaint submission time: under 90 seconds for a text-only complaint.
- Real-time notification latency: consistently under 1 second from admin update to teacher notification.
- Excel import performance: 500-row batch insert completed in under 3 seconds.
- RLS enforcement: zero unauthorised data access observed in penetration testing scenarios.
- Category routing accuracy: 100% in all tested cases across all four trade categories.

The super-admin analytics module accurately computed resolution counts and average resolution times for each administrator. The system was demonstrated to five teachers and two administrators in a controlled walkthrough — all participants found the interface intuitive and required less than five minutes of orientation before independent operation.

24. Conclusion

This paper has presented the design, development, and evaluation of the Seven-Eleven School Complaint Management System. The system addresses a well-defined and practically significant gap in existing school information management tools — the absence of a structured, digital, role-aware complaint lifecycle management mechanism.

By combining a teacher-facing complaint submission interface with a trade-categorised admin routing system and a super-admin analytics dashboard, the project delivers a complete lifecycle management solution for school maintenance complaints. The use of Supabase provides real-time capability, row-level security, and relational data integrity without the overhead of dedicated server management.

The system was validated through a comprehensive testing regime and demonstrated consistently correct behaviour across all defined test cases. The results indicate that the system can meaningfully reduce complaint resolution time, improve administrative accountability, and provide data-driven insights to school management.

25. Future Scope

- AI-Based Categorisation: Implement a natural language processing model to automatically classify complaint categories from the description text.
- Native Mobile Application: Develop dedicated iOS and Android apps using React Native or Flutter for push notifications and native camera integration.
- Escalation Engine: Add automatic escalation for complaints unaddressed beyond a configurable SLA threshold.
- HRMS Integration: Link admin performance data with the school's Human Resource Management System for formal performance reviews.
- Multi-School Support: Extend to a multi-tenant model for school district management across multiple campuses.
- Voice Complaint Submission: Add speech-to-text capability to improve accessibility for teaching staff.
- Predictive Maintenance: Analyse historical complaint data to predict recurring failures and enable proactive maintenance scheduling.

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